AI-Based Authentic Face Image Classification and Verification System

Capstone Project Report — LaunchED Global Internship (Data Analytics)

Name : SIPRA MISHRA

1. Problem Statement

Fake and AI-generated face images are increasingly used to create fraudulent social media profiles, misinformation, and manipulated media. This project develops a Machine Learning / Deep Learning model that analyzes facial images and classifies them as Authentic (REAL) or Manipulated (FAKE), using image attributes such as facial characteristics, demographic information, image quality, and metadata. The goal is to support future integration with broader deepfake detection systems.

2. Dataset Description

The dataset contained 6,557 labeled face images with associated metadata, including classification labels, gender, age group, image quality, resolution, confidence scores, detection difficulty, and dataset split (train/validation/test).

- REAL images: 2,790 (sourced from Unsplash)
- FAKE images: 3,767 (generated using StyleGAN3)

For training efficiency, a balanced subset of 2,000 images (1,000 REAL, 1,000 FAKE) was sampled, of which 1,758 were successfully downloaded and used for model training and evaluation.

3. Tools and Technologies Used

- Python, Google Colab (GPU runtime)
- PyTorch and Torchvision — pretrained ResNet50 for feature extraction
- Scikit-learn — Logistic Regression classifier, evaluation metrics
- Pandas, NumPy — data handling
- Gradio — interactive web demo for live predictions

4. Methodology / Workflow

4.1 Data Preprocessing

Images were downloaded from source URLs, converted to RGB, and resized to 224×224 pixels to match the input requirements of the pretrained CNN.

4.2 Feature Extraction

A pretrained ResNet50 (trained on ImageNet) was used as a fixed feature extractor by removing its final classification layer, producing a 2048-dimensional feature vector per image. This transfer learning approach was chosen over training a CNN from scratch, since it leverages existing visual knowledge and requires substantially less data and compute.

4.3 Model Training

The extracted features were split into training (80%) and test (20%) sets using stratified sampling. A Logistic Regression classifier was trained on the training features to distinguish REAL from FAKE images.

4.4 Avoiding Data Leakage

During exploratory analysis, metadata fields such as resolution, source, and detection_difficulty were found to correlate almost perfectly with the label. These fields were excluded from the feature set to ensure the model learns from genuine visual patterns rather than metadata shortcuts.

5. Results

5.1 Evaluation Metrics (Test Set)

Metric	Value
Accuracy	97.44%
Precision	98.97%
Recall	96.50%
F1-Score	97.72%

5.2 Confusion Matrix

Actual \ Predicted	Predicted FAKE	Predicted REAL
Actual FAKE	150	2
Actual REAL	7	193

Out of 152 actual FAKE images, 150 were correctly identified (only 2 missed). Out of 200 actual REAL images, 193 were correctly identified (7 misclassified).

6. Demographic and Fairness Analysis

As required by the project objectives, model performance was further analyzed across gender, age group, and image quality to check for bias:

Group	Accuracy
Gender: Female	99.72%
Gender: Male	98.88%
Gender: Unknown	99.62%
Age 18-25	99.32%
Age 26-35	99.55%
Age 36-50	99.34%
Age 50+	99.76%
Image Quality: High	99.49%
Image Quality: Medium	99.47%

The model performed consistently well across all demographic groups and quality levels, with accuracy remaining above 98.8% in every category, indicating no significant demographic bias in predictions.

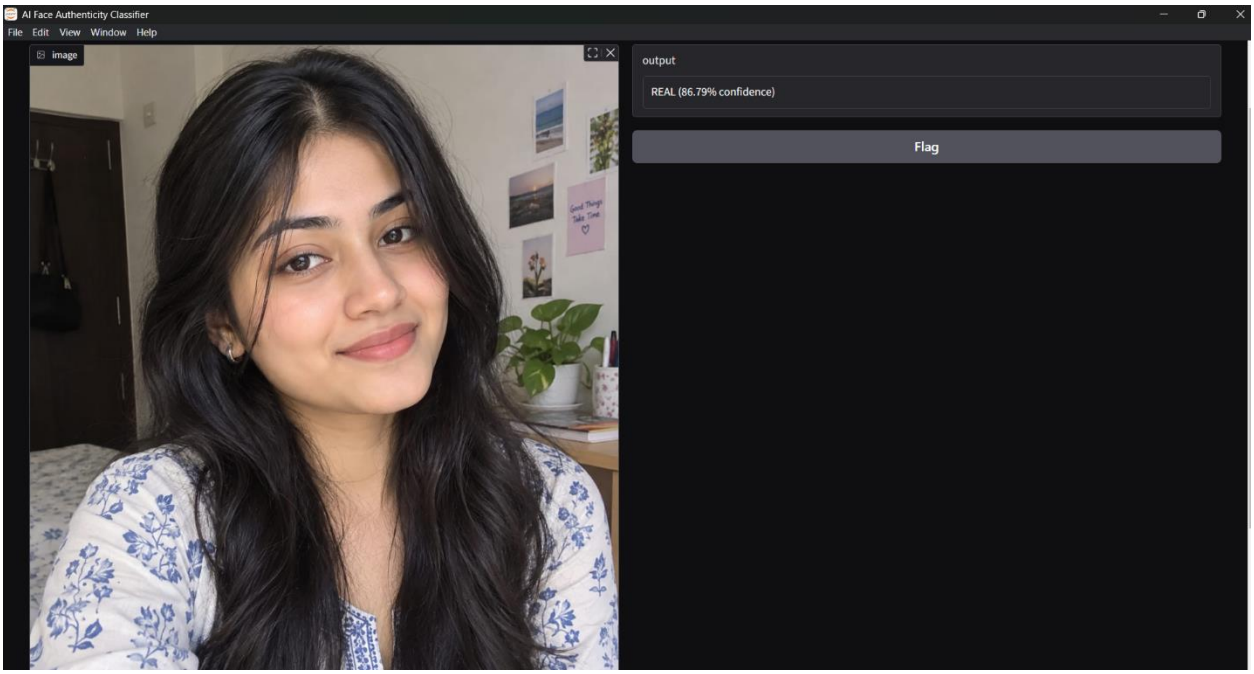
7. Live Demonstration

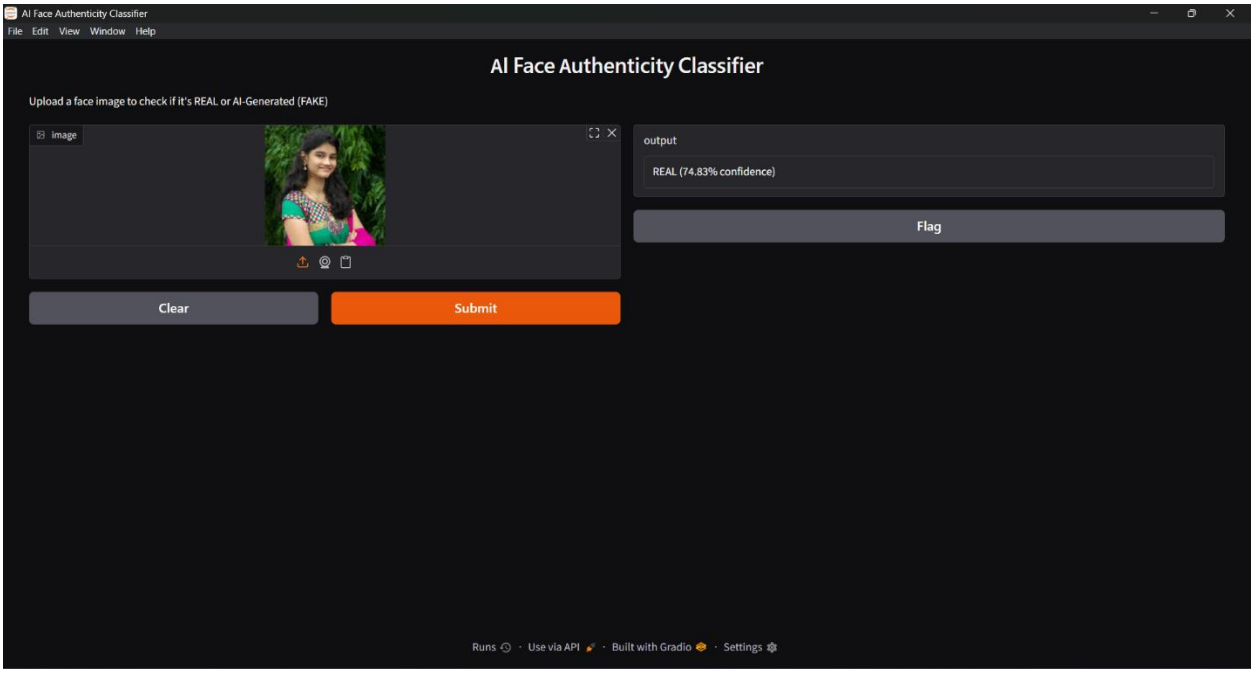
A Gradio-based web application was built to demonstrate the model interactively. Users can upload a face image and receive an instant REAL / FAKE prediction with a confidence score. The app was tested successfully on real photographs.

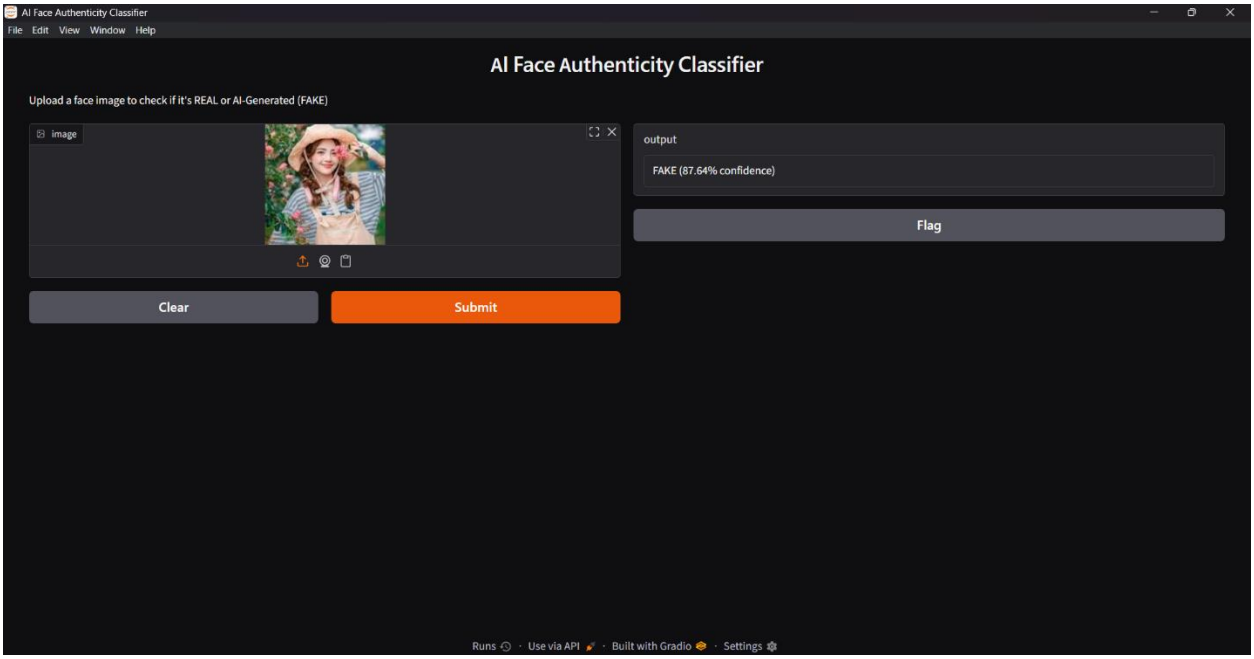
8. Challenges Faced and Solutions

- Class imbalance between REAL and FAKE images — addressed by sampling a balanced subset for training.
- Potential data leakage through metadata fields (resolution, source) — identified during EDA and excluded from the feature set.
- Generalization limits — the model, trained only on StyleGAN3-generated fakes, was found to misclassify faces generated by other methods (e.g., diffusion-based models) as REAL. This reflects a known challenge in deepfake detection: classifiers often do not generalize well across different generation techniques, and would benefit from training on a more diverse set of fake-image sources.
- The model is also scoped specifically to face images; non-face inputs (illustrations, objects, animals) fall outside its intended use case.

9. Results/Output Screenshots







10. Conclusion and Future Work

The developed system successfully classifies facial images as authentic or AI-generated with high accuracy (97.44%) and strong fairness across demographic groups, demonstrating the practical application of transfer learning and computer vision techniques in digital image verification. Future improvements could include training on a more diverse dataset spanning multiple fake-generation techniques (GANs and diffusion models), fine-tuning the CNN backbone instead of using it as a fixed feature extractor, and extending the system toward broader deepfake detection use cases.